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d and f Block Elements

NEET Chemistry · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	106 questions
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MCQ (Single Correct Answer) 106 questions

Q1. RE-NEET 2026

The lanthanide ion having four unpaired electrons is (Given : Atomic numbers of $\mathrm{Ce} = 58, \mathrm{Nd} = 60, \mathrm{Tb} = 65$ and $\mathrm{Ho} = 67$)

(A) Ho^{3+} (B) Nd^{3+} (C) Ce^{3+} (D) Tb^{3+}

Open solution

Q2. RE-NEET 2026

The green paramagnetic species formed by heating $\mathrm{KMnO_4}$ at 513 K is

(A) $\mathrm{KO_2}$ (B) $\mathrm{K_2MnO_4}$ (C) $\mathrm{Mn_3O_4}$ (D) MnO

Open solution

Q3. RE-NEET 2026

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Generally, $3d$ transition metals have high melting points.

Reason R: Involvement of $3d$-electrons in addition to $4s$-electrons in the interatomic metallic bonding.

In light of the above statements, choose the most appropriate answer from the options given below:

(A) $\mathbf{A}$ is not correct but $\mathbf{R}$ is correct

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are correct and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(C) Both $\mathbf{A}$ and $\mathbf{R}$ are correct and $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(D) $\mathbf{A}$ is correct but $\mathbf{R}$ is not correct.

Open solution

Q4. NEET 2026

Match List I with List II :

List I (Transition metal/compound/ complex)		List II (Catalytic Role)	
A.	$\mathrm{V_2O_5}$	I.	Preparation of ammonia from $\mathrm{N_2/H_2}$ mixture
B.	Fe	II.	Polymerisation of alkynes
C.	$\mathrm{PdCl_2}$	III.	Preparation of $\mathrm{H_2SO_4}$ and $\mathrm{SO_2}$
D.	Ni complex	IV.	Oxidation of ethyne to ethanal

Choose the correct answer from the options given below.

(A) A-III, B-IV, C-I, D-II (B) A-IV, B-I, C-III, D-II

(C) A-II, B-I, C-IV, D-III (D) A-III, B-I, C-IV, D-II

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Q5. NEET 2026

Although +3 oxidation state is most common in lanthanoids, cerium still shows +4 oxidation state because:

(A) After losing one more electron, it acquires $4f^{14}$ electronic configuration.

(B) Its nearest inert gas is Radon.

(C) Its atomic number is 61 .

(D) After losing one more electron, it acquires $4f^{0}$ electronic configuration.

Open solution

Q6. NEET 2026

The calculated 'spin-only' magnetic moment $\mathrm{Ti}^{2+}\left(3\,\mathrm{d}^{2}\right)$ is :

(A) 5.92 BM (B) 3.87 BM (C) 2.84 BM (D) 4.90 BM

Open solution

Q7. NEET 2025

Match List - I with List - II.

	List - I		List - II
A	Haber process	I	Fe catalyst
B	Wacker oxidation	II	$\mathrm{PdCl_2}$
C	Wilkinson catalyst	III	$[(\mathrm{PPh_3})_3\mathrm{RhCl}]$
D	Ziegler catalyst	IV	$\mathrm{TiCl_4}$ with $\mathrm{Al(CH_3)_3}$

Choose the correct answer from the options given below :

(A) A-I, B-II, C-III, D-IV (B) A-I, B-IV, C-III, D-II

(C) A-I, B-II, C-IV, D-III (D) A-II, B-III, C-I, D-IV

Open solution

Q8. NEET 2025

Given below are two statements :

Statement I: Ferromagnetism is considered as an extreme form of paramagnetism.

Statement II : The number of unpaired electrons in a Cr^{2+} ion $(\mathrm{Z}=24)$ is the same as that of a Nd^{3+} ion $(\mathrm{Z}=60)$.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

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Q9. NEET 2024 (Re-Examination)

Baeyer's reagent is :

(A) Acidic potassium permanganate solution

(B) Acidic potassium dichromate solution

(C) Cold, dilute, aqueous solution of potassium permanganate

(D) Hot, concentrated solution of potassium permanganate

Open solution

Q10. NEET 2024 (Re-Examination)

Which of the following pairs of ions will have same spin only magnetic moment values within the pair?

A. Zn^{2+}, Ti^{2+}

B. Cr^{2+}, Fe^{2+}

C. Ti^{3+}, Cu^{2+}

D. V^{2+}, Cu^{+}

Choose the correct answer from the options given below :

(A) C and D only (B) A and D only (C) A and B only (D) B and C only

Open solution

Q11. NEET 2024 (Re-Examination)

Which of the following set of ions act as oxidising agents?

(A) Ce^{4+} and Tb^{4+} (B) La^{3+} and Lu^{3+}

(C) Eu^{2+} and Yb^{2+} (D) Eu^{2+} and Tb^{4+}

Open solution

Q12. NEET 2024 (Re-Examination)

The UV-visible absorption bands in the spectra of lanthanoid ions are 'X', probably because of the excitation of electrons involving 'Y'. The 'X' and 'Y', respectively, are :

(A) Broad and f orbitals (B) Narrow and f orbitals

(C) Broad and d and f orbitals (D) Narrow and d and f orbitals

Open solution

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Q13. NEET 2024 (Re-Examination)

Open solution

Match List I with List II:

	List - I (Block/group in periodic table)		List - II (Element)
(A)	Lanthanoid	(I)	Ce
(B)	d-block element	(II)	As
(C)	p-block element	(III)	Cs
(D)	s-block element	(IV)	Mn

Choose the correct answer from the options given below:

(A) A-I, B-II, C-IV, D-III
(B) A-I, B-IV, C-III, D-II
(C) A-I, B-IV, C-II, D-III
(D) A-IV, B-I, C-II, D-III

Q14. NEET 2024 (Re-Examination)

Open solution

Identify the incorrect statement.

(A) PEt_3 and $AsPh_3$ as ligands can form $d\pi - d\pi$ bond with transition metals
(B) The $N - N$ single bond is as strong as the $P - P$ single bond
(C) Nitrogen has unique ability to form $p\pi - p\pi$ multiple bonds with nitrogen, carbon and oxygen
(D) Nitrogen cannot form $d\pi - p\pi$ bond as other heavier elements of its group

Q15. NEET 2024

Open solution

'Spin only' magnetic moment is same for which of the following ions?

A. Ti^{3+}
B. Cr^{2+}
C. Mn^{2+}
D. Fe^{2+}
E. Sc^{3+}

Choose the most appropriate answer from the options given below.

(A) B and D only **(B)** A and E only **(C)** B and C only **(D)** A and D only

Q16. NEET 2024

Open solution

The E° value for the Mn^{3+}/Mn^{2+} couple is more positive than that of Cr^{3+}/Cr^{2+} or Fe^{3+}/Fe^{2+} due to change of

(A) d^5 to d^4 configuration
(B) d^5 to d^2 configuration
(C) d^4 to d^5 configuration
(D) d^3 to d^5 configuration

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Q17. NEET 2024

The pair of lanthanoid ions which are diamagnetic is

(A) Ce^{4+} and Yb^{2+}

(B) Ce^{3+} and Eu^{2+}

(C) Gd^{3+} and Eu^{3+}

(D) Pm^{3+} and Sm^{3+}

Open solution

Q18. NEET 2023 Manipur

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Ionisation enthalpy increases along each series of the transition elements from left to right. However, small variations occur.

Reason (R) : There is corresponding increase in nuclear charge which accompanies the filling of electrons in the inner d-orbitals.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) (A) is correct but (R) is not correct.

(B) (A) is not correct but (R) is correct.

(C) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(D) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

Open solution

Q19. NEET 2023 Manipur

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Ionisation enthalpies of early actinoids are lower than for early lanthanoids.

Reason (R) : Electrons are entering 5f orbitals in actinoids which experience greater shielding from nuclear charge.

In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is true but (R) is false.

(B) (A) is false but (R) is true.

(C) Both (A) and (R) are true and (R) is the correct explanation of (A).

(D) Both (A) and (R) are true but (R) is not the correct explanation of (A).

Open solution

Q20. NEET 2023

The stability of Cu^{2+} is more than Cu^{+} salts in aqueous solution due to -

(A) enthalpy of atomization.

(B) hydration energy.

(C) second ionisation enthalpy.

(D) first ionisation enthalpy.

Open solution

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Q21. NEET 2023

Open solution

Which of the following statements are INCORRECT?

A. All the transition metals except scandium form MO oxides which are ionic.

B. The highest oxidation number corresponding to the group number in transition metal oxides is attained in Sc_2O_3 to Mn_2O_7.

C. Basic character increases from V_2O_3 to V_2O_4 to V_2O_5.

D. V_2O_4 dissolves in acids to give VO_4^{3-} salts.

E. CrO is basic but Cr_2O_3 is amphoteric.

Choose the correct answer from the options given below:

(A) B and D only (B) C and D only (C) B and C only (D) A and E only

Q22. NEET 2022 Phase 2

Open solution

Given below are two statements :

Statement I : Cr^{2+} is oxidising and Mn^{3+} is reducing in nature.

Statement II : Sc^{3+} compounds are repelled by the applied magnetic field.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

Q23. NEET 2021

Open solution

The incorrect statement among the following is :

(A) Actinoids are highly reactive metals, especially when finely divided.

(B) Actinoid contraction is greater for element to element than lanthanoid contraction.

(C) Most of the trivalent Lanthanoid ions are colorless in the solid state.

(D) Lanthanoids are good conductors of heat and electricity.

Q24. NEET 2021

Open solution

Which of the following reactions is the metal displacement reaction? Choose the right option.

(A) $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2 \uparrow$

(B) $2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2$

(C) $Cr_2O_3 + 2Al \xrightarrow{\Delta} Al_2O_3 + 2Cr$

(D) $Fe + 2HCl \rightarrow FeCl_2 + H_2 \uparrow$

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Q25. NEET 2021

Zr (Z = 40) and Hf (Z = 72) have similar atomic and ionic radii because of :

(A) Having similar chemical properties
(B) Belonging to same group
(C) Diagonal relationship
(D) Lanthanoid contraction

Open solution

Q26. NEET 2020 Phase 2

Identify the **incorrect** statement from the following :

(A) The overall decrease in atomic and ionic radii from lanthanum to lutetium is called lanthanoid contraction.
(B) Zirconium and Hafnium have identical radii of 160 pm and 159 pm, respectively as a consequence of lanthanoid contraction.
(C) Lanthanoids reveal only +3 oxidation state.
(D) The lanthanoid ions other than the f^0 type and the f^{14} type are all paramagnetic.

Open solution

Q27. NEET 2020 Phase 2

Match the following aspects with the respective metal.

Aspects	Metal
(a) The metal which reveals a maximum number of oxidation states	(i) Scandium
(b) The metal although placed in 3d block is considered not as a transition element	(ii) Copper
(c) The metal which does not exhibit variable oxidation states	(iii) Manganese
(d) The metal which in +1 oxidation state in aqueous solution undergoes disproportionation	(iv) Zinc

Select the correct option :

(A) (a)-(ii) (b)-(iv) (c)-(i) (d)-(iii)
(B) (a)-(i) (b)-(iv) (c)-(ii) (d)-(iii)
(C) (a)-(iii) (b)-(iv) (c)-(i) (d)-(ii)
(D) (a)-(iii) (b)-(i) (c)-(iv) (d)-(ii)

Open solution

Q28. NEET 2020 Phase 1

Identify the incorrect statement.

(A) The transition metals and their compounds are known for their catalytic activity due to their ability to adopt multiple oxidation states and to form complexes.
(B) Interstitial compounds are those that are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals.
(C) The oxidation states of chromium in CrO_4^{2-} and $Cr_2O_7^{2-}$ are not the same.
(D) Cr^{2+} (d^4) is a stronger reducing agent than Fe^{2+} (d^6) in water.

Open solution

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Q29. NEET 2019 (Odisha)

Open solution

Match the catalyst with the process

Catalyst	Process
(i) V_2O_5	(a) The oxidation of ethyne to ethanal
(ii) $TiCl_4 + Al(CH_3)_3$	(b) Polymerisation of alkynes
(iii) $PdCl_2$	(c) Oxidation of SO_2 in the manufacture of H_2SO_4
(iv) Nickel complexes	(d) Polymerisation of ethylene

Which of the following is the correct option?

(A) (i)-(c), (ii)-(a), (iii)-(d), (iv)-(b)
(B) (i)-(c), (ii)-(d), (iii)-(a), (iv)-(b)
(C) (i)-(a), (ii)-(b), (iii)-(c), (iv)-(d)
(D) (i)-(a), (ii)-(c), (iii)-(b), (iv)-(d)

Q30. NEET 2019 (Odisha)

Open solution

When neutral or faintly alkaline $KMnO_4$ is treated with potassium iodide, iodide ion is converted into 'X'. 'X' is

(A) IO^- **(B)** I_2 **(C)** IO_4^- **(D)** IO_3^-

Q31. NEET 2018

Open solution

Which one of the following ions exhibits d-d transition and paramagnetism as well?

(A) CrO_4^{2-} **(B)** $Cr_2O_7^{2-}$ **(C)** MnO_4^- **(D)** MnO_4^{2-}

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Q32. NEET 2018

Open solution

Match the metal ions given in Column I with the spin magnetic moments of the ions given in Column II and assign the correct code :

Column - I	Column - II
A. Co^{3+}	(i) $\sqrt{8}$ B.M.
B. Cr^{3+}	(ii) $\sqrt{35}$ B.M.
C. Fe^{3+}	(iii) $\sqrt{3}$ B.M.
D. Ni^{2+}	(iv) $\sqrt{24}$ B.M.
	(v) $\sqrt{15}$ B.M.

(A)

A	B	C	D
(iv)	(v)	(ii)	(i)

(B)

A	B	C	D
(i)	(ii)	(iii)	(iv)

(C)

A	B	C	D
(iv)	(i)	(ii)	(iii)

(D)

A	B	C	D
(iii)	(v)	(i)	(ii)

Q33. NEET 2017

Open solution

Name the gas that can readily decolourise acidified $KMnO_4$ solution.

(A) SO_2 (B) NO_2 (C) P_2O_5 (D) CO_2

Q34. NEET 2017

Open solution

$HgCl_2$ and I_2 both when dissolved in water containing I^- ions, the pair of species formed is

(A) HgI_2, I^- (B) HgI_4^{2-}, I_3^- (C) Hg_2I_2, I^- (D) HgI_2, I_3^-

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Q35. NEET 2017

The reason for greater range of oxidation states in actinoids is attributed to

(A) actinoid contraction

(B) $5f$, 6d and 7s levels having comparable energies

(C) $4f$ and 5d levels being close in energies

(D) the radioactive nature of actinoids.

Open solution

Q36. NEET 2016 Phase 2

Which one of the following statements related to lanthanoids is incorrect?

(A) Europium shows + 2 oxidation state.

(B) The basicity decreases as the ionic radius decreases from Pr to Lu.

(C) All the lanthanons are much more reactive than aluminium.

(D) Ce(+4) solutions are widely used as oxidizing agent in volumetric analysis.

Open solution

Q37. NEET 2016 Phase 1

Which one of the following statements is correct when SO_2 is passed through acidified $K_2Cr_2O_7$ solution?

(A) SO_2 is reduced.

(B) Green $Cr_2(SO_4)_3$ is formed.

(C) The solution turns blue.

(D) The solution is decolourised.

Open solution

Q38. NEET 2016 Phase 1

The electronic configuration of Eu (Atomic No. 63), Gd (Atomic No. 64) and Tb (Atomic No.65) are

(A) $[Xe]4f^6 5d^1 6s^2$, $[Xe]4f^7 5d^1 6s^2$ and $[Xe]4f^8 5d^1 6s^2$

(B) $[Xe]4f^7\ 6s^2$, $[Xe]4f^7\ 5d^1\ 6s^2$ and $[Xe]4f^9\ 6s^2$

(C) $[Xe]4f^7\ 6s^2$, $[Xe]4f^8\ 6s^2$ and $[Xe]4f^8\ 5d^1\ 6s^2$

(D) $[Xe]4f^6\ 5d^1\ 6s^2$, $[Xe]4f^7\ 5d^1\ 6s^2$ and $[Xe]4f^9\ 6s^2$

Open solution

Q39. AIPMT 2015

Gadolinium belongs to 4f series. Its atomic number is 64. Which of the following is the correct electronic configuration of gadolinium?

(A) $[Xe]4f^9 5s^1$ (B) $[Xe]4f^7 5d^1 6s^2$ (C) $[Xe]4f^6 5d^2 6s^2$ (D) $[Xe]4f^8 6d^2$

Open solution

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Q40. AIPMT 2015

Assuming complete ionisation, same moles of which of the following compounds will require the least amount of acidified $KMnO_4$ for complete oxidation?

(A) $FeSO_3$ (B) FeC_2O_4 (C) $Fe(NO_2)_2$ (D) $FeSO_4$

Open solution

Q41. AIPMT 2015 Cancelled Paper

Magnetic moment 2.84 B.M. is given by (At. nos. Ni = 28, Ti = 22, Cr = 24, Co = 27)

(A) Cr^{2+} (B) Co^{2+} (C) Ni^{2+} (D) Ti^{3+}

Open solution

Q42. AIPMT 2015 Cancelled Paper

Which of the following processes does not involve oxidation of iron?

(A) Formation of $Fe(CO)_5$ from Fe.

(B) Liberation of H_2 from steam by iron at high temperature.

(C) Rusting of iron sheets.

(D) Decolourisation of blue $CuSO_4$ solution by iron.

Open solution

Q43. AIPMT 2015 Cancelled Paper

Because of lanthanoid contraction, which of the following pairs of elements have nearly same atomic radii? (Numbers in the parenthesis are atomic numbers)

(A) Zr(40) and Hf(72) (B) Zr(40) and Ta(73)

(C) Ti(22) and Zr(40) (D) Zr(40) and Nb(41)

Open solution

Q44. AIPMT 2014

The reaction of aqueous $KMnO_4$ with H_2O_2 in acidic conditions gives

(A) Mn^{4+} and O_2 (B) Mn^{2+} and O_2 (C) Mn^{2+} and O_3 (D) Mn^{4+} and MnO_2

Open solution

Q45. AIPMT 2014

Magnetic moment 2.83 BM is given by which of the following ions?
(At. nos. Ti = 22, Cr = 24, Mn = 25, Ni = 28)

(A) Ti^{3+} (B) Ni^{2+} (C) Cr^{3+} (D) Mn^{2+}

Open solution

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Q46. AIPMT 2014

Reason of lanthanoid contraction is

(A) negligible screening effect of 'f'- orbitals

(B) increasing nuclear charge

(C) decreasing nuclear charge

(D) decreasing screening effect.

Open solution

Q47. NEET 2013 (Karnataka)

Sc (Z = 21) is a transition element but Zn (Z = 30) is not because

(A) both Sc^{3+} and Zn^{2+} ions are colourless and form white compounds.

(B) in case of Sc, 3d orbitals are partially filled but in Zn these are filled.

(C) last electron is assumed to be added to 4s level in case of Zn.

(D) both Sc and Zn do not exhibit variable oxidation states.

Open solution

Q48. NEET 2013

Which of the following statements about the interstitial compounds is incorrect?

(A) They are much harder than the pure metal.

(B) They have higher melting points than the pure metal.

(C) The retain metallic conductivity.

(D) They are chemically reactive.

Open solution

Q49. NEET 2013

Which of the following lanthanoid ions is diamagnetic?
(At. nos. Ce = 58, Sm = 62, Eu = 63, Yb = 70)

(A) Eu^{2+} (B) Yb^{2+} (C) Ce^{2+} (D) Sm^{2+}

Open solution

Q50. AIPMT 2012 Mains

The catalytic activity of transition metals and their compounds is ascribed mainly to

(A) their magnetic behaviour

(B) their unfilled d-orbitals

(C) their ability to adopt variable oxidation states

(D) their chemical reactivity

Open solution

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Q51. AIPMT 2012 Mains

Which of the following exhibits only + 3 oxidation state?

(A) U (B) Th (C) Ac (D) Pa

Open solution

Q52. AIPMT 2012 Mains

Which one of the following does not correctly represent the correct order of the property indicated against it?

(A) Ti < V < Cr < Mn; increasing number of oxidation states

(B) $Ti^{3+} < V^{3+} < Cr^{3+} < Mn^{3+}$: increasing magnetic moment

(C) Ti < V < Cr < Mn : increasing melting points

(D) Ti < V < Mn < Cr : increasing 2nd ionization enthalpy

Open solution

Q53. AIPMT 2012 Mains

Four successive members of the first series of the transition metals are listed below. For which one of them the standard potential $\left(E^{o}_{M^{2+}/M}\right)$ value has a positive sign?

(A) Co (Z = 27) (B) Ni (Z = 28) (C) Cu (Z = 29) (D) Fe (Z = 26)

Open solution

Q54. AIPMT 2012 Prelims

Identify the alloy containing a non-metal as a constituent in it.

(A) Invar (B) Steel (C) Bell metal (D) Bronze

Open solution

Q55. AIPMT 2012 Prelims

Which of the statements is not true?

(A) On passing H_2S through acidified $K_2Cr_2O_7$ solution, a milky colour is observed.

(B) $Na_2Cr_2O_7$ is preferred over $K_2Cr_2O_7$ in volumetric analysis.

(C) $K_2Cr_2O_7$ solution in acidic medium is orange.

(D) $K_2Cr_2O_7$ solution becomes yellow increasing the pH beyond 7.

Open solution

Q56. AIPMT 2011 Prelims

For the four successive transition elements (Cr, Mn, Fe and Co), the stability of + 2 oxidation state will be there in which of the following order?
(At. nos. Cr = 24, Mn = 25, Fe = 26, Co = 27)

(A) Mn > Fe > Cr > Co (B) Fe > Mn > Co > Cr

(C) Co > Mn > Fe > Cr (D) Cr > Mn > Co > Fe

Open solution

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Q57. AIPMT 2011 Prelims

Acidified $K_2Cr_2O_7$ solution turns green when Na_2SO_3 is added to it. This is due to the formation of

(A) $Cr_2(SO_4)_3$ (B) CrO_4^{2-} (C) $Cr_2(SO_3)_3$ (D) $CrSO_4$

Open solution

Q58. AIPMT 2010 Mains

Match List I (substances) with List II (processes) employed in the manufacture of the substances and select the correct option.

Column I (Substances)			Column II (Processes)	
(A)	Sulphuric acid		(i)	Haber's process
(B)	Steel		(ii)	Bessemer's process
(C)	Sodium hydroxide		(iii)	Leblanc process
(D)	Ammonia		(iv)	Contact process

(A) (A) → (i); (B) → (iv); (C) → (ii); (D) → (iii)

(B) (A) → (i); (B) → (ii); (C) → (iii); (D) → (iv)

(C) (A) → (iv); (B) → (iii); (C) → (ii); (D) → (i)

(D) (A) → (iv); (B) → (ii); (C) → (iii); (D) → (i)

Open solution

Q59. AIPMT 2010 Mains

Which of the following oxidation states is the most common among the lanthanoids?

(A) 4 (B) 2 (C) 5 (D) 3

Open solution

Q60. AIPMT 2010 Prelims

Which of the following ions will exhibit colour in aqueous solutions?

(A) La^{3+} (Z = 57) (B) Ti^{3+} (Z = 22) (C) Lu^{3+} (Z = 71) (D) Sc^{3+} (Z = 21)

Open solution

Q61. AIPMT 2010 Prelims

Which of the following ions has electronic configuration $[Ar]3d^6$?
(At. nos. Mn = 25, Fe = 26, Co = 27, Ni = 28)

(A) Ni^{3+} (B) Mn^{3+} (C) Fe^{3+} (D) Co^{3+}

Open solution

Q62. AIPMT 2010 Prelims

Which of the following pairs has the same size?

(A) Fe^{2+}, Ni^{2+} (B) Zr^{4+}, Ti^{4+} (C) Zr^{4+}, Hf^{4+} (D) Zn^{2+}, Hf^{4+}

Open solution

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Q63. AIPMT 2009

Which one of the elements with the following outer orbital configurations may exhibit the largest number of oxidation states?

(A) $3d^5 4s^1$ (B) $3d^5 4s^2$ (C) $3d^2 4s^2$ (D) $3d^3 4s^2$

Open solution

Q64. AIPMT 2008

The correct order of decreasing second ionisation enthalpy of Ti(22), V(23), Cr(24) and Mn(25) is

(A) Mn > Cr > Ti > V (B) Ti > V > Cr > Mn (C) Cr > Mn > V > Ti (D) V > Mn > Cr > Ti

Open solution

Q65. AIPMT 2007

Which one of the following ions is the most stable in aqueous solution?
(At. No. Ti = 22, V = 23, Cr = 24, Mn = 25)

(A) V^{3+} (B) Ti^{3+} (C) Mn^{3+} (D) Cr^{3+}

Open solution

Q66. AIPMT 2007

Identify the incorrect statement among the following:

(A) Lanthanoid contraction is the accumulation of successive shrinkages.

(B) As a result of lanthanoid contraction, the properties of 4d series of the transition elements have no similarities with the 5d series of elements.

(C) Shielding power of $4f$ electrons is quite weak.

(D) There is a decrease in the radii of the atoms or ions as one proceeds from La to Lu.

Open solution

Q67. AIPMT 2006

In which of the following pairs are both the ions coloured in aqueous solution?
(At. no. : Sc = 21, Ti = 22, Ni = 28, Cu = 29, Co = 27)

(A) Ni^{2+}, Cu^{+} (B) Ni^{2+}, Ti^{3+} (C) Sc^{3+}, Ti^{3+} (D) Sc^{3+}, Co^{2+}.

Open solution

Q68. AIPMT 2006

Copper sulphate dissolves in excess of KCN to give

(A) $Cu(CN)_2$ (B) CuCN (C) $[Cu(CN)_4]^{3-}$ (D) $[Cu(CN)_4]^{2-}$

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Q69. AIPMT 2006

More number of oxidation states are exhibited by the actinoids than by the lanthanoids. The main reason for this is

(A) more active nature of the actinoids

(B) more energy difference between 5f and 6d orbitals than that between 4f and 5d orbitals

(C) lesser energy difference between 5f and 6d orbitals than that between 4f and 5d orbitals

(D) greater metallic character of the lanthanoids than that of the corresponding actinoids.

Open solution

Q70. AIPMT 2005

More number of oxidation states are exhibited by the actinoids than by the lanthanoids. The main reason for this is

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(B) more energy difference between 5f and 6d orbitals than that between 4f and 5d orbitals

(C) lesser energy difference between 5f and 6d orbitals than that between 4f and 5d orbitals

(D) greater metallic character of the lanthanoids than that of the corresponding actinoids.

Open solution

Q71. AIPMT 2005

The number of moles of $KMnO_4$ reduced by one mole of KI in alkaline medium is

(A) one (B) two (C) five (D) one fifth

Open solution

Q72. AIPMT 2005

Four successive members of the first row transition elements are listed below with their atomic numbers. Which one of them is expected to have the highest third ionisation enthalpy?

(A) Vanadium (Z = 23) (B) Chromium (Z = 24)

(C) Manganese (Z = 25) (D) Iron (Z = 26)

Open solution

Q73. AIPMT 2005

The aqueous solution containing which one of the following ions will be colourless?
(Atomic number : Sc = 21, Fe = 26, Ti = 22, Mn = 25)

(A) Sc^{3+} (B) Fe^{2+} (C) Ti^{3+} (D) Mn^{2+}

Open solution

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Q74. AIPMT 2004

Among the following series of transition metal ions, the one where all metal ions have $3d^2$ electronic configuration is
[At. Nos. Ti = 22, V = 23, Cr = 24, Mn = 25]

Open solution

(A) $Ti^{3+}, V^{2+}, Cr^{3+}, Mn^{4+}$
(B) $Ti^{+}, V^{4+}, Cr^{6+}, Mn^{7+}$
(C) $Ti^{4+}, V^{3+}, Cr^{2+}, Mn^{3+}$
(D) $Ti^{2+}, V^{3+}, Cr^{4+}, Mn^{5+}$

Q75. AIPMT 2004

lanthanoids are

Open solution

(A) 14 elements in the sixth period (atomic no. 90 to 103) that are filling $4f$ sublevel
(B) 14 elements in the seventh period (atomic number = 90 to 103) that are filling $5f$ sublevel
(C) 14 elements in the sixth period (atomic number = 58 to 71) that are filling the $4f$ sublevel
(D) 14 elements in the seventh period (atomic number = 50 to 71) that are filling $4f$ sublevel.

Q76. AIPMT 2003

Which one of the following characteristics of the transition metals is associated with their catalytic activity?

Open solution

(A) High enthalpy of atomization
(B) Paramagnetic behaviour
(C) Colour of hydrated ions
(D) Variable oxidation states

Q77. AIPMT 2003

The basic character of the transition metal monoxides follows the order
(Atomic no's. Ti = 22, V = 23, Cr = 24, Fe = 26)

Open solution

(A) VO > CrO > TiO > FeO
(B) CrO > VO > FeO > TiO
(C) TiO > FeO > VO > CrO
(D) TiO > VO > CrO > FeO

Q78. AIPMT 2003

The correct order of ionic radii of Y^{3+}, La^{3+}, Eu^{3+} and Lu^{3+} is
(Atomic nos. Y = 39, La = 57, Eu = 63, Lu = 71)

Open solution

(A) $Y^{3+} < La^{3+} < Eu^{3+} < Lu^{3+}$
(B) $Y^{3+} < Lu^{3+} < Eu^{3+} < La^{3+}$
(C) $Lu^{3+} < Eu^{3+} < La^{3+} < Y^{3+}$
(D) $La^{3+} < Eu^{3+} < Lu^{3+} < Y^{3+}$

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Q79. AIPMT 2002

General electronic configuration of lanthanides is

(A) $(n-2)f^{1-14}\ (n-1)s^2p^6d^{0-1}\ ns^2$

(B) $(n-2)f^{0-14}\ (n-1)d^{0-1}\ ns^2$

(C) $(n-2)f^{0-14}\ (n-1)d^{10}\ ns^2$

(D) $(n-2)d^{0-1}\ (n-1)\ f^{1-14}\ ns^2$

Open solution

Q80. AIPMT 2002

An atom has electronic configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3\ 4s^2$, you will place it in which group?

(A) fifth (B) fifteenth (C) second (D) third.

Open solution

Q81. AIPMT 2002

Which of the following shows maximum number of oxidation states?

(A) Cr (B) Fe (C) Mn (D) V

Open solution

Q82. AIPMT 2002

Zn gives H_2 gas with H_2SO_4 and HCl but not with HNO_3 because

(A) Zn act as oxidising agent when react with HNO_3

(B) HNO_3 is weaker acid than H_2SO_4 and HCl

(C) In electrochemical series Zn is above hydrogen

(D) NO_3^- is reduced in preference to hydronium ion.

Open solution

Q83. AIPMT 2001

Which of the following statement is **not** correct?

(A) $La(OH)_3$ is less basic than $Lu(OH)_3$.

(B) In lanthanide series ionic radius of Ln^{+3} ion decreases.

(C) La is actually an element of transition series rather than lanthanides.

(D) Atomic radius of Zr and Hf are same because of lanthanide contraction.

Open solution

Q84. AIPMT 2001

The most convenient method to protect the bottom of ship made of iron is

(A) coating it with red lead oxide (B) white tin plating

(C) connecting it with Mg block (D) connecting it with Pb block.

Open solution

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Q85. AIPMT 2000

Which of the following shows maximum number of oxidation states?

(A) Cr (B) Fe (C) Mn (D) V

Open solution

Q86. AIPMT 2000

Which ion is colourless?

(A) Cr^{4+} (B) Sc^{3+} (C) Ti^{3+} (D) V^{3+}

Open solution

Q87. AIPMT 1999

Which of the following has more unpaired d-electrons?

(A) Zn^{+} (B) Fe^{2+} (C) N^{3+} (D) Cu^{+}

Open solution

Q88. AIPMT 1999

Bell-metal is an alloy of

(A) $Cu + Pb$ (B) $Cu + Sn$ (C) $Cu + Zn$ (D) $Cu + Ni$

Open solution

Q89. AIPMT 1998

Which one of the following ionic species will impart colour to an aqueous solution?

(A) Ti^{4+} (B) Cu^{+} (C) Zn^{2+} (D) Cr^{3+}

Open solution

Q90. AIPMT 1998

Which one of the following elements shows maximum number of different oxidation states in its compounds?

(A) Eu (B) La (C) Gd (D) Am

Open solution

Q91. AIPMT 1997

Which of the following does not represent the correct order of the property indicated?

(A) $Mn^{2+} > Ni^{2+} < Co^{2+} < Fe^{2+}$ (B) $Fe^{2+} > Co^{2+} > Ni^{2+} > Cu^{2+}$

(C) $Ni^{2+} > Cr^{2+} > Fe^{2+} > Mn^{2+}$ (D) $Se > Ti > Cr > Mn$

Open solution

Q92. AIPMT 1997

The lanthanide contraction is responsible for the fact that

(A) Zr and Hf have about the same radius (B) Zr and Zn have the same oxidation state

(C) Zr and Y have about the same radius (D) Zr and Nb have similar oxidation state.

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Q93. AIPMT 1997

Which of the following element is responsible for oxidation of water to O_2 in biological processes?

(A) Cu (B) Mo (C) Fe (D) Mn

Open solution

Q94. AIPMT 1997

The electronic configuration of gadolinium (Atomic No. = 64) is

(A) $[Xe]\,4f^3\,5d^5\,6s^2$ (B) $[Xe]\,4f^6\,5d^2\,6d^2$ (C) $[Xe]\,4f^8\,5d^9\,6s^2$ (D) $[Xe]\,4f^7\,5d^1\,6s^2$

Open solution

Q95. AIPMT 1997

$K_2Cr_2O_7$ on heating with aqueous NaOH gives

(A) $Cr_2O_7^{2-}$ (B) $Cr(OH)_2$ (C) CrO_4^{2-} (D) $Cr(OH)_3$

Open solution

Q96. AIPMT 1996

A transition element X has a configuration $[Ar]\,3d^4$ in its +3 oxidation state. Its atomic number is

(A) 22 (B) 19 (C) 25 (D) 26

Open solution

Q97. AIPMT 1996

When calomel reacts with NH_4OH, we get

(A) Hg_2O (B) HgO (C) $HgNH_2Cl$ (D) $NH_2-Hg-Hg-Cl$

Open solution

Q98. AIPMT 1996

The electronic configuration of transition elements is exhibited by

(A) ns^1 (B) ns^2np^5 (C) $ns^2\,(n-1)d^{1-10}$ (D) $ns^2\,(n-1)d^{10}$

Open solution

Q99. AIPMT 1995

Amongst TiF_6^{2-}, CoF_6^{3-}, Cu_2Cl_2 and $NiCl_4^{2-}$, which are the colourless species? (atomic number of Ti = 22, Co = 27, Cu = 29, Ni = 28)

(A) CoF_6^{3-} and $NiCl_4^{2-}$ (B) TiF_6^{2-} and Cu_2Cl_2
(C) Cu_2Cl_2 and $NiCl_4^{2-}$ (D) TiF_6^{2-} and CoF_6^{3-}

Open solution

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Q100. AIPMT 1995

The mercury is the only metal which is liquid at 0°C. This is due to its

(A) high vapour pressure (B) weak metallic bond

(C) high ionization energy (D) both (b) and (c).

Open solution

Q101. AIPMT 1994

Which of the following shows maximum number of oxidation states?

(A) Cr (B) Fe (C) Mn (D) V

Open solution

Q102. AIPMT 1994

Which of the following statement concerning lanthanide elements is false?

(A) All lanthanides are highly dense metals.

(B) More characteristic oxidation state of lanthanide elements is +3.

(C) Lanthanides are separated from one another by ion exchange method.

(D) Ionic radii of trivalent lanthanides steadily increases with increase in the atomic number.

Open solution

Q103. AIPMT 1991

The transition elements have a general electronic configuration

(A) $ns^2np^6nd^{1-10}$ (B) $(n-1)d^{1-10}$, ns^{0-2}, np^{0-6}

(C) $(n-1)d^{1-10}$, ns^{1-2} (D) $nd^{1-10}\,ns^{-2}$

Open solution

Q104. AIPMT 1989

Photographic films and plates have an essential ingredient of

(A) silver nitrate (B) silver bromide (C) sodium chloride (D) oleic acid.

Open solution

Q105. AIPMT 1989

Nitriding is the process of surface hardening of steel by treating it is an atmosphere of

(A) NH_3 (B) O_3 (C) N_2 (D) H_2S.

Open solution

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Q106. AIPMT 1989

A blue colouration is not obtained when

(A) ammonium hydroxide dissolves in copper sulphate

(B) copper sulphate solution reacts with $K_4[Fe(CN)_6]$

(C) ferric chloride reacts with sod. ferrocyanide

(D) anhydrous $CuSO_4$ is dissolved in water.

Open solution

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Answer Key

1. **A**	2. **B**	3. **B**	4. **D**	5. **D**
6. **C**	7. **A**	8. **A**	9. **C**	10. **D**
11. **A**	12. **B**	13. **C**	14. **B**	15. **A**
16. **C**	17. **A**	18. **C**	19. **C**	20. **B**
21. **B**	22. **A**	23. **C**	24. **C**	25. **D**
26. **C**	27. **C**	28. **C**	29. **B**	30. **D**
31. **D**	32. **A**	33. **A**	34. **B**	35. **B**
36. **C**	37. **B**	38. **B**	39. **B**	40. **D**
41. **C**	42. **A**	43. **A**	44. **B**	45. **B**
46. **A**	47. **B**	48. **D**	49. **B**	50. **C**
51. **C**	52. **C**	53. **C**	54. **B**	55. **B**
56. **A**	57. **A**	58. **D**	59. **D**	60. **B**
61. **D**	62. **C**	63. **B**	64. **C**	65. **C**
66. **B**	67. **B**	68. **C**	69. **C**	70. **C**
71. **B**	72. **C**	73. **A**	74. **D**	75. **C**
76. **D**	77. **D**	78. **B**	79. **A**	80. **A**
81. **C**	82. **D**	83. **A**	84. **B**	85. **C**
86. **B**	87. **B**	88. **B**	89. **D**	90. **D**
91. **C**	92. **A**	93. **D**	94. **D**	95. **C**
96. **C**	97. **C**	98. **C**	99. **B**	100. **D**
101. **C**	102. **D**	103. **C**	104. **B**	105. **A**
106. **B**				

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